

US Credit Card Complaints Analysis Project

Table of Contents

1. [Project Overview](#)
2. [Data Overview](#)
3. [Objectives](#)
4. [SQL and Tableau Calculated Fields](#)
 - [SQL Queries](#)
 - [Tableau Calculated Fields](#)
5. [Tableau Visualization](#)
6. [Insights and Recommendations](#)
7. [Conclusion](#)
8. [Additional Resources](#)

Project Overview

This project focuses on analyzing credit card complaints data, offering insights into common issues, response times, and geographic distribution. By leveraging Tableau, the analysis provides a visual representation of complaint trends, helping stakeholders address customer pain points more effectively.

Data Overview

- Credit Card Complaints Data: This dataset includes complaints logged against various financial institutions, categorized by issues like billing disputes, identity theft, and others. It also includes details on company responses and the channels through which complaints were submitted.

Objectives

- Complaint Analysis: Identify the most common types of complaints and track their frequency over time.

- Response Analysis: Evaluate the efficiency of the companies in responding to complaints.
- Geographical Insights: Analyze the distribution of complaints across different regions.
- Customer Satisfaction: Examine the correlation between the type of response and customer satisfaction.

SQL and Tableau Calculated Fields

SQL Queries

Here are some example SQL queries used in this project:

```
CREATE DATABASE IF NOT EXISTS complaints_db
  CHARACTER SET = 'utf8mb4'
  COLLATE = 'utf8mb4_unicode_ci';

USE complaints_db;

CREATE TABLE IF NOT EXISTS complaints (
  complaint_id          VARCHAR(255) NOT NULL PRIMARY KEY,
  complaint_id_alt      VARCHAR(255),
  company               VARCHAR(255) NOT NULL,
  company_public_response TEXT,
  company_response_to_consumer TEXT,
  complaint_category    VARCHAR(200),
  consumer_complaint_narrative TEXT,
  consumer_consent_provided TINYINT(1) DEFAULT NULL,      -- 1 = true, 0 =
false, NULL = unknown
  consumer_consent_group VARCHAR(100),
  consumer_disputed      TINYINT(1) DEFAULT NULL,      -- 1 = true, 0 =
false
  date_received          DATE,
  date_sent_to_company2  DATE,
  dimension              VARCHAR(200),
  issue                  VARCHAR(200),
  product                VARCHAR(200),
  select_company_or_state VARCHAR(100),
  state                  VARCHAR(100),
  sub_issue               VARCHAR(200),
  sub_product            VARCHAR(200),
  submitted_via           VARCHAR(100),
  source_table_name      VARCHAR(255),
  tags                    JSON,                          -- JSON array of
tags, e.g. '["tag1","tag2"]'
  timely_response        TINYINT(1) DEFAULT NULL,
  zip_code               VARCHAR(20),
  avg_no_of_days         DECIMAL(10,2),
```

```

    donut                                JSON,                                -- store chart
data/metadata here (or change to DECIMAL if numeric)
    number_of_records                    INT,
    created_at                           DATETIME DEFAULT CURRENT_TIMESTAMP,
    updated_at                           DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP
) ENGINE=InnoDB
  DEFAULT CHARSET=utf8mb4
  COLLATE=utf8mb4_unicode_ci;

-- Indexes for faster lookups
CREATE INDEX idx_company ON complaints(company);
CREATE INDEX idx_product ON complaints(product);
CREATE INDEX idx_date_received ON complaints(date_received);
CREATE INDEX idx_state ON complaints(state);

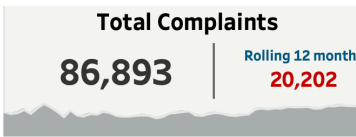
-- Load Data
BULK INSERT ComplaintsData
FROM 'path_to_your/credit_card_data.csv'
WITH (FIELDTERMINATOR = ',', ROWTERMINATOR = '\n', FIRSTROW = 2);

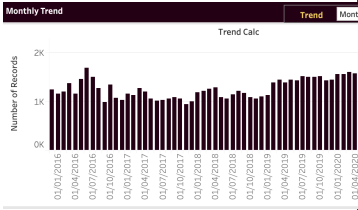
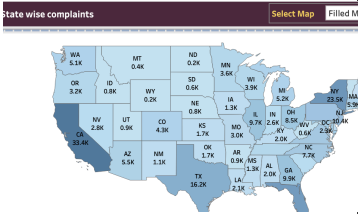
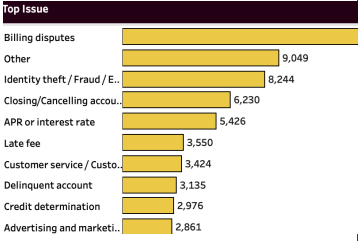
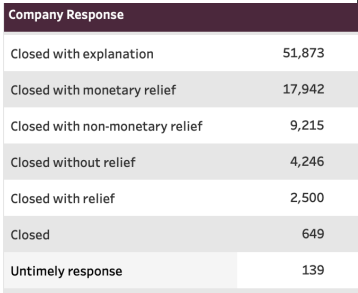
```

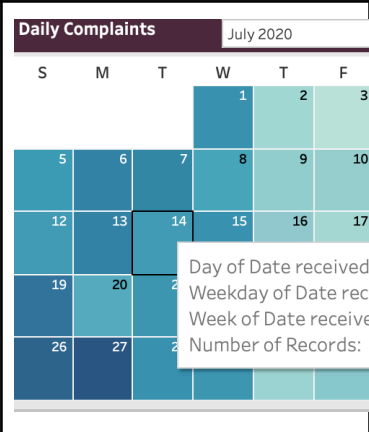
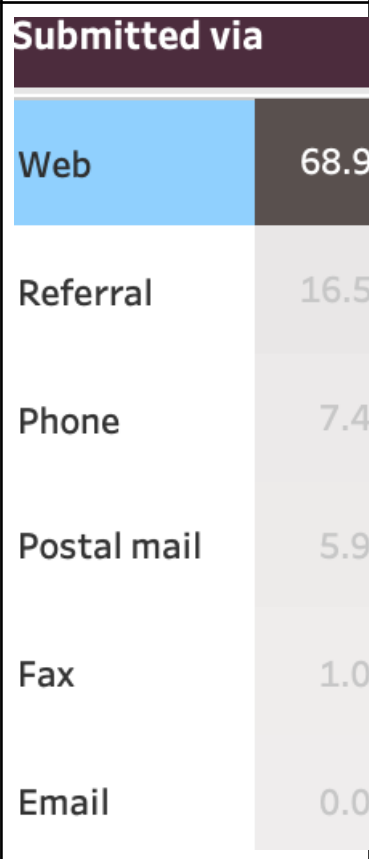
Data Cleaning

1. Handle Duplicates & IDs
2. Standardize Text Fields
3. Handle Missing / Null Values
4. Validate Dates & Timelines
5. Numeric Fields
6. Data Type Conversions
 - Convert Yes/No columns → Boolean (1/0).
 - ZIP codes → keep as VARCHAR (leading zeros important).
 - Dates → convert all to DATE type.
7. Outlier & Consistency Checks

Dashboard charts and KPIs

Visual Card	Description	Calculation
	KPI: Total Complaints and Last 12-month: This KPI presents the total number of credit card complaints received, along with the rolling value for the last twelve months, highlighting recent trends in complaint volume.	- SUM([Rolling 12 month complaints]) { FIXED : sum(IF [Rolling 12 month]= TRUE then ([Number of Records]) END)} (DATEDIFF('month', DATETRUNC('month',[Date received]), DATETRUNC('month',[Maximun Date Parameter]))<12
	KPI: In Progress & % of In-progress Complaints: This KPI tracks the number of complaints that are currently in progress, indicating the workload of customer service teams, along with the percentage of total complaints that are still unresolved.	- SUM([Timely Response])/SUM([Number of Records]) - IF [Timely response?]='Yes' then 1 else 0 END
	KPI: Timely Response & % of Closed Complaints: This metric measures the proportion of closed complaints that received a timely response within a predefined timeframe, providing insights into customer service efficiency.	SUM(IF [Company response to consumer1]='In Progress' THEN [Number of Records] end)/SUM([Number of Records])

	Chart: Weekly Trend of Complaints: This visualization displays the weekly trend of credit card complaints over time, including an average line to identify patterns, and highlights the maximum value of complaints reported in a single week.	DATE((DATETRUNC([Trend],[Date received]))))
	Chart: State-wise Complaints Density Map: This chart showcases the geographical distribution of credit card complaints across different states, enabling stakeholders to identify regions with high complaint densities.	SUM([Number of Records])
	Chart: Top Issue-wise Complaints: This visualization presents the top issues reported by customers, offering insights into the most common grievances faced by credit card users	SUM([Number of Records])
	Company Response by Total Complaints and % of Total: This chart illustrates the response rate of credit card companies to complaints, both in terms of the total number of complaints and the percentage of total complaints responded to.	SUM([Number of Records])

	Daily Complaints Heat Map in the Form of Calendar Chart: This visualization represents the daily distribution of credit card complaints using a calendar chart, allowing stakeholders to identify days with high complaint volumes	SUM([Number of Records])
	Submitted via Total Complaints Chart: This chart categorizes complaints based on the channels through which they were submitted (e.g., phone, email, website), providing insights into customer communication preferences and channel effectiveness.	SUM([Number of Records])

Insights and Recommendations

Based on the analysis:

- Identify theft/ Fraud/Embezzlement: These are the most common complaints, indicating a need for improved billing processes.
- Geographical Focus: States like California and Texas have higher complaint rates, suggesting targeted customer service improvements in these areas.
- Response Efficiency: While most complaints are closed with an explanation, a higher focus on providing monetary relief might improve customer satisfaction.

- **Submission Channels:** A majority of complaints are submitted online, highlighting the importance of a user-friendly web interface for complaint submission.

Conclusion

This analysis provides critical insights into credit card complaint patterns, helping companies identify key areas for improvement in customer service and complaint resolution.